

HARDY Z AS A TIME-CHANGED WEAKLY STATIONARY GAUSSIAN PROCESS

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APPENDIX A. THE HARDY Z -FUNCTION AS A WEAKLY HARMONIZABLE TIME-WARPED STATIONARY GAUSSIAN PROCESS

This appendix develops, in self-contained form, the second-order stochastic classification of the Hardy Z -function under the spectral representation derived in the main text. It establishes that Z is a centered Gaussian process of the weakly harmonizable, time-warped stationary class (equivalently $G(\theta)$ -stationary, reducibly-to-weakly-stationary), specifies the exact spectral density $S(\xi)$ as a convergent sum of stationary-phase Gaussians plus an endpoint atom, derives the exact autocovariance in fully substituted closed form, and extracts the variance structure function from the diagonal.

A.1. Setting and notation. Fix $T_0 \geq 200$ throughout and let

$$\begin{aligned} (1) \quad & \theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \\ (2) \quad & \theta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \\ (3) \quad & \theta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right), \\ (4) \quad & Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right). \end{aligned}$$

Lemma A.1 (Binet bound). *For every $t \geq 200$,*

$$(5) \quad \left| \theta'(t) - \frac{1}{2} \log(t/(2\pi)) \right| \leq \frac{1}{200},$$

and $\theta''(t) > 0$ on $[200, \infty)$.

Proof. Binet's first formula for $\operatorname{Re} \psi$ applied with $s = \frac{1}{4} + \frac{it}{2}$ gives $\operatorname{Re} \psi(s) = \log |s| - \operatorname{Re} \int_0^\infty (e^u - 1)^{-1} \cdot u/(u^2 + 4|s|^2) du - \frac{1}{2} \operatorname{Re}(1/s)$. Using $|s|^2 = \frac{1}{16} + (t/2)^2$ and $\operatorname{Re} s = \frac{1}{4}$, one has $\log |s| = \log(t/2) + \frac{1}{2} \log(1 + \frac{1}{4t^2})$. For $t \geq 200$, $\frac{1}{2} \log(1 + \frac{1}{4t^2}) \leq 3.2 \cdot 10^{-6}$, the Binet remainder integral is bounded by $\frac{1}{12|s|^2} \leq \frac{1}{3t^2}$, and $|\frac{1}{2} \operatorname{Re}(1/s)| \leq \frac{1}{2t^2}$. Summing, $|\frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2} \log(t/2)| \leq 1/(2.4 \cdot 10^4) < 1/200$. Subtracting $\frac{1}{2} \log \pi$ yields (5). For θ'' , $\psi'(s) = \sum_{k \geq 0} (s+k)^{-2}$, each term with $\operatorname{Im}(s+k)^{-2} = -2(\frac{1}{4} + k)(t/2)/|s+k|^4 < 0$, hence $\theta''(t) > 0$. $\square$

By Lemma A.1, θ is a C^∞ -diffeomorphism of $[T_0, \infty)$ onto $[\theta(T_0), \infty)$, and θ' is a C^∞ -diffeomorphism of $[T_0, \infty)$ onto $[\theta'(T_0), \infty)$. Denote the inverses by θ^{-1} and $(\theta')^{-1}$.

Let H be the process on the θ -axis defined by

$$(6) \quad H(u) := \frac{Z(\theta^{-1}(u))}{\sqrt{\theta'(\theta^{-1}(u))}}, \quad u \in [\theta(T_0), \infty),$$

equivalently

$$(7) \quad Z(t) = \sqrt{\theta'(t)} H(\theta(t)).$$

The main text establishes that H admits a Cramér spectral representation

$$(8) \quad H(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi), \quad \mathbb{E}[d\Phi(\xi) \overline{d\Phi(\eta)}] = \delta(\xi - \eta) S(\xi) d\xi,$$

with orthogonal complex Gaussian spectral measure $d\Phi$ and density $S(\xi) \geq 0$ supported in $[-1, 1]$. Set

$$(9) \quad \sigma_H^2 := \int_{-1}^1 S(\xi) d\xi = \mathbb{E}|H(u)|^2,$$

a finite positive constant.

A.2. Exact spectral density. For $n \in \mathbb{Z}_{\geq 2}$ and $\sigma \in \{+1, -1\}$, the stationary-phase locus of mode (n, σ) is

$$(10) \quad t_{n,\sigma}^*(\xi) := (\theta')^{-1}\left(\frac{\log n}{\sigma - \xi}\right), \quad \xi \in \sigma \cdot (0, 1),$$

and the mode amplitude squared is

$$(11) \quad |\mathcal{A}_{n,\sigma}(\xi)|^2 = \frac{2\pi \log n}{n(\sigma - \xi)^2 \theta''(t_{n,\sigma}^*(\xi))}.$$

Let $w \geq 0$ denote the Gabcke remainder weight of the main text, concentrated at $\xi = \pm 1$. Then

$$(12) \quad S(\xi) = \frac{1}{(2\pi)^2} \sum_{\sigma \in \{-1, +1\}} \sum_{n=2}^{\infty} |\mathcal{A}_{n,\sigma}(\xi)|^2 \mathbf{1}_{\sigma \cdot (0,1)}(\xi) + w[\delta(\xi - 1) + \delta(\xi + 1)].$$

A.3. Classification theorem.

Theorem A.2 (Classification). *The process Z defined by (4) is a centered Gaussian process on $[T_0, \infty)$ belonging to the following classes simultaneously:*

- (1) Reducibly weakly stationary (RWS): H in (6) is weakly stationary on $[\theta(T_0), \infty)$ with reducing diffeomorphism θ ([3, Ch. 4], [6]).
- (2) $G(\theta)$ -stationary in the sense of Gray–Vijverberg–Woodward:

$$(13) \quad Z(t) = \sqrt{\theta'(t)} \int_{-1}^1 e^{i\xi\theta(t)} d\Phi(\xi)$$

([6]).

- (3) Deformed (time-warped) stationary in the sense of Clerc–Mallat: $Z = D_\theta H$ with $(D_\theta f)(t) := \sqrt{\theta'(t)} f(\theta(t))$ unitary on $L^2(\mathbb{R})$ ([5]).
- (4) Weakly harmonizable in the sense of Rozanov–Rao, with spectral bimeasure of finite Fréchet variation and infinite Vitali variation ([4, 2]).

Furthermore, Z lies outside the Priestley oscillatory class and outside the Chang–Rao oscillatory harmonizable class.

Proof. Gaussianity. $d\Phi$ is complex Gaussian in (8), so H is centered Gaussian, and $Z(t)$ is a deterministic multiple of H at a deterministic argument.

(1) *RWS.* From (6), the covariance $\mathbb{E}[H(u_1)\overline{H(u_2)}] = \int e^{i\xi(u_1-u_2)} S(\xi) d\xi$ by (8), hence depends only on $u_1 - u_2$.

(2) *$G(\theta)$ -stationary.* Substitute (8) into (7).

(3) *Deformed stationary.* For $f \in L^2(\mathbb{R})$, the substitution $u = \theta(t)$, $du = \theta'(t) dt$ gives $\int |(D_\theta f)(t)|^2 dt = \int \theta'(t) |f(\theta(t))|^2 dt = \int |f(u)|^2 du$.

(4) *Weak harmonizability.* By Theorem A.6, $\text{Var } Z(t) = \sigma_H^2 \theta'(t) \rightarrow \infty$, ruling out strong harmonizability (which requires bounded variance). The bimeasure of Z is the pushforward under the diffeomorphism θ of the orthogonal measure $d\Phi$ on $[-1, 1]$, an orthogonally scattered measure of finite total mass σ_H^2 . By [4, Thm. 1], every orthogonally scattered random measure induces a bimeasure of finite Fréchet variation equal to its total mass.

Exclusion from Priestley and Chang–Rao. Writing (13) formally against the rigid kernel $e^{i\lambda t}$ with $\lambda = \xi$ gives amplitude envelope $A_t(\lambda) = \sqrt{\theta'(t)} e^{i\lambda[\theta(t)-t]}$. Consider the exact Fourier transform

$F_\lambda(\vartheta) := \int_{T_0}^\infty A_t(\lambda) e^{-i\vartheta t} dt$ with total phase $\Phi(t; \vartheta, \lambda) = \lambda[\theta(t) - t] - \vartheta t$. The stationary point of Φ satisfies $\theta'(t^*) = 1 + \vartheta/\lambda$. By Lemma A.1, θ' is a strictly increasing surjection $[T_0, \infty) \rightarrow [\theta'(T_0), \infty)$, so there is a unique stationary point $t^* = t^*(\vartheta, \lambda) \in [T_0, \infty)$ whenever $1 + \vartheta/\lambda \geq \theta'(T_0)$. Differentiating $t^*(\vartheta, \lambda)$ with respect to ϑ at fixed $\lambda > 0$ gives $\partial t^*/\partial \vartheta = 1/(\lambda \theta''(t^*)) > 0$, so t^* is strictly increasing in ϑ and unbounded above as $\vartheta \rightarrow \infty$. The exact stationary-phase amplitude $|F_\lambda(\vartheta)| = \sqrt{2\pi \theta'(t^*)/|\lambda \theta''(t^*)|} = \sqrt{2\pi(1 + \vartheta/\lambda)/(\lambda \theta''(t^*))}$ is strictly increasing in ϑ for $\lambda > 0$, since both $1 + \vartheta/\lambda$ increases and $\theta''(t^*) \rightarrow 0$ as $t^* \rightarrow \infty$ (by $\theta''(t) = O(1/t)$). Hence $|F_\lambda(\vartheta)|$ has no maximum on $\mathbb{R}$, failing both Priestley's origin-concentration condition and Chang–Rao's existence-of-maximum condition. $\square$

A.4. Exact autocovariance.

Theorem A.3 (Exact autocovariance, spectral-integral form). *For all $t, s \in [T_0, \infty)$,*

$$(14) \quad R_Z(t, s) := \mathbb{E}[Z(t)\overline{Z(s)}] = \sqrt{\theta'(t)\theta'(s)} \int_{-1}^1 e^{i\xi[\theta(t)-\theta(s)]} S(\xi) d\xi.$$

Proof. By (7) and (8), $\mathbb{E}[Z(t)\overline{Z(s)}] = \sqrt{\theta'(t)\theta'(s)} \mathbb{E}\left[\int e^{i\xi\theta(t)} d\Phi(\xi) \int e^{-i\eta\theta(s)} d\overline{\Phi}(\eta)\right]$. The orthogonality $\mathbb{E}[d\Phi(\xi)d\overline{\Phi}(\eta)] = \delta(\xi - \eta)S(\xi)d\xi$ reduces the double integral to (14). $\square$

Theorem A.4 (Exact autocovariance, fully substituted closed form). *With $\Delta := \theta(t) - \theta(s)$,*

$$(15) \quad R_Z(t, s) = \frac{2\sqrt{\theta'(t)\theta'(s)}}{\pi} \sum_{n=2}^{\infty} \frac{\log n}{n} \int_0^1 \frac{\cos((1-v)\Delta)}{v^2 \theta''((\theta')^{-1}(\log n/v))} dv + 2w\sqrt{\theta'(t)\theta'(s)} \cos \Delta.$$

Proof. Substitute (12) into (14). The continuous part contributes, for each $\sigma \in \{+1, -1\}$ and $n \geq 2$,

$$I_{n,\sigma} := \frac{1}{(2\pi)^2} \int_{\sigma \cdot (0,1)} e^{i\xi\Delta} \frac{2\pi \log n}{n(\sigma - \xi)^2 \theta''(t_{n,\sigma}^*(\xi))} d\xi.$$

Substitute $v := \sigma - \xi$, so $\xi = \sigma - v$ and $d\xi = -dv$. When $\sigma = +1$, $\xi \in (0, 1)$ corresponds to $v \in (0, 1)$ with $\xi = 1 - v$; when $\sigma = -1$, $\xi \in (-1, 0)$ corresponds to $v \in (0, 1)$ with $\xi = -1 + v$ (equivalently $\xi = -(1 - v)$). The integration limits reverse sign but so does $d\xi$, so the net effect is that both σ -integrals become $\int_0^1 \dots dv$ without sign change. Then for $\sigma = +1$: $e^{i\xi\Delta} = e^{i(1-v)\Delta}$ and $(\sigma - \xi)^2 = v^2$, $t_{n,+}^*(\xi) = (\theta')^{-1}(\log n/v)$. For $\sigma = -1$: $e^{i\xi\Delta} = e^{-i(1-v)\Delta}$ and $(\sigma - \xi)^2 = v^2$, $t_{n,-}^*(\xi) = (\theta')^{-1}(\log n/v)$ (same function of v). Therefore

$$I_{n,+} + I_{n,-} = \frac{\log n}{2\pi n} \int_0^1 \frac{e^{i(1-v)\Delta} + e^{-i(1-v)\Delta}}{v^2 \theta''((\theta')^{-1}(\log n/v))} dv = \frac{\log n}{\pi n} \int_0^1 \frac{\cos((1-v)\Delta)}{v^2 \theta''((\theta')^{-1}(\log n/v))} dv.$$

Summing over $n \geq 2$ and multiplying by $\sqrt{\theta'(t)\theta'(s)}$ from (14) yields the continuous part of (15). The atomic part $w[\delta(\xi - 1) + \delta(\xi + 1)]$ contributes $w\sqrt{\theta'(t)\theta'(s)}(e^{i\Delta} + e^{-i\Delta}) = 2w\sqrt{\theta'(t)\theta'(s)} \cos \Delta$. $\square$

Corollary A.5 (Sample-path inner-product form). *For $[T_0, T]$ and any lag η ,*

$$(16) \quad C_H(\eta) = \int_{T_0}^{T_\eta} Z(\tau) Z(\theta^{-1}(\theta(\tau) + \eta)) \sqrt{\frac{\theta'(\tau)}{\theta'(\theta^{-1}(\theta(\tau) + \eta))}} d\tau,$$

with $T_\eta = \theta^{-1}(\theta(T) - \eta)$. At zero lag,

$$(17) \quad C_H(0) = \int_{T_0}^T |Z(\tau)|^2 d\tau = \int_{T_0}^T \left| \zeta\left(\frac{1}{2} + i\tau\right) \right|^2 d\tau.$$

Proof. $C_H(\eta) = \int_{\theta(T_0)}^{\theta(T)-\eta} H(u)H(u + \eta) du$; substitute $u = \theta(\tau)$, $du = \theta'(\tau) d\tau$, and apply (6). Identity (17) uses $|Z| = |\zeta(\frac{1}{2} + i\cdot)|$. $\square$

A.5. Variance structure function.

Theorem A.6 (Variance structure function). *For every $t \geq T_0 \geq 200$,*

$$(18) \quad \text{Var } Z(t) = \sigma_H^2 \theta'(t),$$

and consequently

$$(19) \quad \left| \text{Var } Z(t) - \frac{\sigma_H^2}{2} \log(t/(2\pi)) \right| \leq \frac{\sigma_H^2}{200}.$$

Proof. Setting $s = t$ in (14) gives $\Delta = 0$, hence the exponential equals 1, and $\text{Var } Z(t) = \theta'(t) \int_{-1}^1 S(\xi) d\xi = \sigma_H^2 \theta'(t)$ by (9). Multiplying the Binet bound (5) by σ_H^2 gives (19). $\square$

Corollary A.7 (Consistency with the closed form). *Setting $s = t$ in (15) gives $\Delta = 0$, so all cosines equal 1:*

$$(20) \quad \text{Var } Z(t) = \frac{2\theta'(t)}{\pi} \sum_{n=2}^{\infty} \frac{\log n}{n} \int_0^1 \frac{dv}{v^2 \theta''((\theta')^{-1}(\log n/v))} + 2w\theta'(t),$$

which equals $\sigma_H^2 \theta'(t)$ by (9) applied to (12).

Corollary A.8 (Unbounded variance). *$\text{Var } Z(t) \rightarrow \infty$ as $t \rightarrow \infty$. The variance structure function grows logarithmically and does not converge to a constant, consistent with the non-stationarity of Z .*

A.6. Unifying representation.

Theorem A.9 (Gelfand–Karhunen–Loève representation). *Let X be a second-order process on $I \subset \mathbb{R}$ with continuous covariance kernel $K(t, s) = \mathbb{E}[X(t)\overline{X(s)}]$. There exist a σ -finite measure space $(\Lambda, \mathcal{B}, \mu)$, kernels $\{\varphi_\lambda(t)\} \subset L_{\text{loc}}^2(I)$, and an orthogonal random measure M with $\mathbb{E}[M(A)\overline{M(B)}] = \mu(A \cap B)$, such that*

$$(21) \quad X(t) = \int_{\Lambda} \varphi_\lambda(t) dM(\lambda), \quad K(t, s) = \int_{\Lambda} \varphi_\lambda(t) \overline{\varphi_\lambda(s)} d\mu(\lambda).$$

Proof. Gelfand–Karhunen–Loève theorem [1, 2]. $\square$

Corollary A.10 (GKL form of Z). *Z admits the GKL representation (21) with $\Lambda = [-1, 1]$, $d\mu(\xi) = S(\xi) d\xi$ given by (12), $\varphi_\xi(t) = \sqrt{\theta'(t)} e^{i\xi\theta(t)}$, and $dM(\xi) = d\Phi(\xi)/\sqrt{S(\xi)}$.*

A.7. Summary of constants.

Quantity	Value	Source
Binet error $ \theta'(t) - \frac{1}{2} \log(t/(2\pi)) $	$\leq 1/200$	Lemma A.1
Spectral density $S(\xi)$	(12)	§A.2
Total spectral mass $\int S$	σ_H^2	(9)
Atom weight at $\xi = \pm 1$	w	(12)
$\text{Var } Z(t)/\theta'(t)$	σ_H^2	Theorem A.6
Variance log-approximation error	$\leq \sigma_H^2/200$	Theorem A.6
Domain of validity	$t \geq 200$	Lemma A.1

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